

Budget Audit Report

1. Cover

Evidence-first engineering budget / composition / BDI audit.

2. Case identification

Case: pr-budget-01
Generated: 2026-08-01T20:00:53Z
Documents: 1

3. Executive summary

Items: 12
Findings: 31 — {'MEDIUM': 27, 'HIGH': 4}
Arithmetic checks: 22 — {'PASS': 17, 'MATERIAL_DIFFERENCE': 2, 'NOT_EVALUATED': 3}
Sum total_sale_price (items with value): 5256.25
Sum total_direct_cost (items with value): None
BDI components: 8; sum_pct_points=30.5

4. Scope and limitations

Human remains responsible for price formation, margin, competitive strategy and seal. Missing formula cache is never treated as zero. BDI is not margin. No win probability or optimal bid.

5. Files and sheets

Document golden-11ff148c: golden.xlsx
sha256=606b43e6fd86bf6f9c4ded28b569742d12abe9c144472bbd4791236cf88b0a86 cells=191

6. Extraction quality

See workbook extraction-quality artifacts. Case phase=audited.

7. Budget summary

Normalized items=12; with sale totals=11; with direct totals=0.

8. Formula integrity

Integrity issue_count=7; formula_issues tracked in audits.

9. Arithmetic reconciliation

Checks=22; status_counts={'PASS': 17, 'MATERIAL_DIFFERENCE': 2, 'NOT_EVALUATED': 3};
policy={'absolute_tolerance_br': 0.01, 'relative_tolerance_pct': 0.05, 'rounding_tolerance': 0.005,
'materiality_br': 100.0, 'materiality_pct_of_total': 0.1}

10. Quantities

Quantity audit: 4 issues

11. Prices

Unit-price audit note: Unit price comparisons require explicit reference manifest

12. Compositions

Compositions=1; inputs=4; issues=14

13. Social charges

Social charges: 3

14. BDI

Components=8; sum_percent_points=30.5; non_claims=['BDI legal', 'BDI ilegal', 'BDI correto', 'BDI abusivo', 'BDI conforme TCU', 'BDI is margin']

- Administração central: raw=5.0 frac=0.05 rule=component_percent_points
- Riscos: raw=1.5 frac=0.015 rule=component_percent_points
- Seguros: raw=0.5 frac=0.005 rule=component_percent_points
- Garantias: raw=0.5 frac=0.005 rule=component_percent_points
- Despesas financeiras: raw=1.0 frac=0.01 rule=component_percent_points
- Tributos: raw=8.0 frac=0.08 rule=component_percent_points
- Lucro: raw=6.0 frac=0.06 rule=component_percent_points
- Tributos: raw=8.0 frac=0.08 rule=component_percent_points

15. Schedule

Schedule reconciliation in audits/ when schedule items present.

16. ABC curve

ABC materiality classes are not error/overprice labels.

17. Official references

References: NOT_RUN

18. Risks

Risk count=21; non_claims=['Does not conclude inexecutable', 'Does not invent internal cost', 'Does not invent win probability', 'Does not suggest optimal bid', 'Does not treat BDI as margin']

19. Prioritized findings

F-0001 [MEDIUM] ARITHMETIC_ERROR: Material arithmetic difference — cells: Orcamento Analitico!E3, Orcamento Analitico!H3, Orcamento Analitico!I3, Orcamento Analitico!F3 — reported=300.0 recomputed=250.0

F-0002 [MEDIUM] ARITHMETIC_ERROR: Material arithmetic difference — cells: Orcamento Analitico!E8, Orcamento Analitico!H8, Orcamento Analitico!I8, Orcamento Analitico!F8 — reported=156.25 recomputed=125.0

F-0003 [HIGH] FORMULA_ERROR: Formula issue: BROKEN_REFERENCE — cells: Orcamento Analitico!I4 — reported=None recomputed=None

F-0004 [MEDIUM] FORMULA_ERROR: Formula issue: EXTERNAL_REFERENCE — cells: Auxiliar!B2 — reported=None recomputed=None

F-0005 [MEDIUM] ARITHMETIC_ERROR: NEGATIVE_VALUE — cells: Orcamento Analitico!A9, Orcamento Analitico!B9, Orcamento Analitico!C9, Orcamento Analitico!D9, Orcamento Analitico!E9, Orcamento Analitico!F9, Orcamento Analitico!H9, Orcamento Analitico!I9, Orcamento Analitico!G9 — reported=None recomputed=None

F-0006 [MEDIUM] ARITHMETIC_ERROR: NEGATIVE_VALUE — cells: Orcamento Analitico!A9, Orcamento Analitico!B9, Orcamento Analitico!C9, Orcamento Analitico!D9, Orcamento Analitico!E9, Orcamento Analitico!F9, Orcamento Analitico!H9, Orcamento Analitico!I9, Orcamento Analitico!G9 — reported=None recomputed=None

F-0007 [MEDIUM] POSSIBLE_DUPLICATION: DUPLICATE_CODE — cells: Orcamento Analitico!A2, Orcamento Analitico!B2, Orcamento Analitico!C2, Orcamento Analitico!D2, Orcamento Analitico!E2, Orcamento Analitico!F2, Orcamento Analitico!H2, Orcamento Analitico!I2, Orcamento Analitico!G2, Orcamento Analitico!A5, Orcamento Analitico!B5, Orcamento Analitico!C5, Orcamento Analitico!D5, Orcamento Analitico!E5, Orcamento Analitico!F5, Orcamento Analitico!H5, Orcamento Analitico!I5, Orcamento Analitico!G5, Curva ABC!A2, Curva ABC!B2, Curva ABC!C2, Curva ABC!D2, Curva ABC!E2 — reported=None recomputed=None

F-0008 [MEDIUM] POSSIBLE_DUPLICATION: DUPLICATE_CODE — cells: Orcamento Analitico!A3, Orcamento Analitico!B3, Orcamento Analitico!C3, Orcamento Analitico!D3, Orcamento Analitico!E3, Orcamento Analitico!F3, Orcamento Analitico!H3, Orcamento Analitico!I3, Orcamento Analitico!G3, Orcamento Analitico!A6, Orcamento Analitico!B6, Orcamento Analitico!C6, Orcamento Analitico!D6, Orcamento Analitico!E6, Orcamento Analitico!F6, Orcamento Analitico!H6, Orcamento Analitico!I6, Orcamento Analitico!G6 — reported=None recomputed=None

F-0009 [MEDIUM] POSSIBLE_DUPLICATION: DUPLICATE_CODE — cells: Orcamento Analitico!A7, Orcamento Analitico!B7, Orcamento Analitico!C7, Orcamento Analitico!D7, Orcamento Analitico!E7, Orcamento Analitico!F7, Orcamento Analitico!H7, Orcamento Analitico!I7, Orcamento Analitico!G7, Curva ABC!A3, Curva ABC!B3, Curva ABC!C3, Curva ABC!D3, Curva ABC!E3 — reported=None recomputed=None

F-0010 [HIGH] QUANTITY_DIVERGENCE: NEGATIVE_QUANTITY — cells: Orcamento Analitico!A9, Orcamento Analitico!B9, Orcamento Analitico!C9, Orcamento Analitico!D9, Orcamento Analitico!E9, Orcamento Analitico!F9, Orcamento Analitico!H9, Orcamento Analitico!I9, Orcamento Analitico!G9 — reported=None recomputed=None

F-0011 [MEDIUM] QUANTITY_DIVERGENCE: ZERO_QUANTITY — cells: Orcamento Analitico!A10, Orcamento Analitico!B10, Orcamento Analitico!C10, Orcamento Analitico!D10, Orcamento Analitico!E10, Orcamento Analitico!F10, Orcamento Analitico!H10, Orcamento Analitico!I10, Orcamento Analitico!G10 — reported=None recomputed=None

F-0012 [MEDIUM] QUANTITY_DIVERGENCE: SAME_CODE_DIFFERENT_QTY — cells: — — reported=None recomputed=None

F-0013 [MEDIUM] QUANTITY_DIVERGENCE: SAME_CODE_DIFFERENT_QTY — cells: — — reported=None recomputed=None

F-0014 [MEDIUM] COMPOSITION_GAP: MISSING_COEFFICIENT — cells: Composicoes!A4, Composicoes!B4, Composicoes!C4, Composicoes!E4 — reported=None recomputed=None

F-0015 [MEDIUM] COMPOSITION_GAP: NEGATIVE_COEFFICIENT — cells: Composicoes!A5, Composicoes!B5, Composicoes!C5, Composicoes!D5, Composicoes!E5, Composicoes!F5 — reported=None recomputed=None

F-0016 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A2, Orcamento Analitico!B2, Orcamento Analitico!C2, Orcamento Analitico!D2, Orcamento Analitico!E2, Orcamento Analitico!F2, Orcamento Analitico!H2, Orcamento Analitico!I2, Orcamento Analitico!G2 — reported=None recomputed=None

F-0017 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A3, Orcamento Analitico!B3, Orcamento Analitico!C3, Orcamento Analitico!D3, Orcamento Analitico!E3, Orcamento Analitico!F3, Orcamento Analitico!H3, Orcamento Analitico!I3, Orcamento Analitico!G3 — reported=None recomputed=None

F-0018 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A4, Orcamento Analitico!B4, Orcamento Analitico!C4, Orcamento Analitico!D4, Orcamento Analitico!E4, Orcamento Analitico!F4, Orcamento Analitico!H4, Orcamento Analitico!I4, Orcamento Analitico!G4 — reported=None recomputed=None

F-0019 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A5, Orcamento Analitico!B5, Orcamento Analitico!C5, Orcamento Analitico!D5, Orcamento Analitico!E5, Orcamento Analitico!F5, Orcamento Analitico!H5, Orcamento Analitico!I5, Orcamento Analitico!G5 — reported=None recomputed=None

F-0020 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A6, Orcamento Analitico!B6, Orcamento Analitico!C6, Orcamento Analitico!D6, Orcamento Analitico!E6, Orcamento Analitico!F6, Orcamento Analitico!H6, Orcamento Analitico!I6, Orcamento Analitico!G6 — reported=None recomputed=None

F-0021 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A7, Orcamento Analitico!B7, Orcamento Analitico!C7, Orcamento Analitico!D7, Orcamento Analitico!E7, Orcamento Analitico!F7, Orcamento Analitico!H7, Orcamento Analitico!I7, Orcamento Analitico!G7 — reported=None recomputed=None

F-0022 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A8, Orcamento Analitico!B8, Orcamento Analitico!C8, Orcamento Analitico!D8, Orcamento Analitico!E8, Orcamento Analitico!F8, Orcamento Analitico!H8, Orcamento Analitico!I8, Orcamento Analitico!G8 — reported=None recomputed=None

F-0023 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A9, Orcamento Analitico!B9, Orcamento Analitico!C9, Orcamento Analitico!D9, Orcamento Analitico!E9, Orcamento Analitico!F9, Orcamento Analitico!H9, Orcamento Analitico!I9, Orcamento Analitico!G9 — reported=None recomputed=None

F-0024 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A10, Orcamento Analitico!B10, Orcamento Analitico!C10, Orcamento Analitico!D10, Orcamento Analitico!E10, Orcamento Analitico!F10, Orcamento Analitico!H10, Orcamento Analitico!I10, Orcamento Analitico!G10 — reported=None recomputed=None

F-0025 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Orcamento Analitico!A11, Orcamento Analitico!B11, Orcamento Analitico!C11, Orcamento Analitico!D11, Orcamento Analitico!E11, Orcamento Analitico!F11, Orcamento Analitico!H11, Orcamento Analitico!I11, Orcamento Analitico!G11 — reported=None recomputed=None

F-0026 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Curva ABC!A2, Curva ABC!B2, Curva ABC!C2, Curva ABC!D2, Curva ABC!E2 — reported=None recomputed=None

F-0027 [MEDIUM] COMPOSITION_GAP: ITEM_WITHOUT_COMPOSITION — cells: Curva ABC!A3, Curva ABC!B3, Curva ABC!C3, Curva ABC!D3, Curva ABC!E3 — reported=None recomputed=None

F-0028 [HIGH] COMPOSITION_GAP: Composition arithmetic material difference — cells: Composicoes!A3, Composicoes!B3, Composicoes!C3, Composicoes!D3, Composicoes!E3, Composicoes!F3 — reported=18.0 recomputed=15.0

F-0029 [MEDIUM] BDI_INCONSISTENCY: DUPLICATE_COMPONENT — cells: — — reported=None recomputed=None

F-0030 [HIGH] BDI_INCONSISTENCY: POSSIBLE_DOUBLE_BDI — cells: Orcamento Analitico!A8, Orcamento Analitico!B8, Orcamento Analitico!C8, Orcamento Analitico!D8, Orcamento Analitico!E8, Orcamento Analitico!F8, Orcamento Analitico!H8, Orcamento Analitico!I8, Orcamento Analitico!G8 — reported=None recomputed=None

F-0031 [MEDIUM] BDI_INCONSISTENCY: COMPONENT_NOT_FOUND_BY_NAME — cells: — — reported=None recomputed=None

20. Human pending items

Review HIGH findings, BDI methodology, and any NEEDS_ENGINEER_REVIEW items.

21. Evidence annex

Primary totals for reconciliation: item_count=12, finding_count=31, total_sale_price_sum=5256.25, total_direct_cost_sum=None, arithmetic_check_count=22.

RECON_FOOTER item_count=12 finding_count=31 total_sale_price_sum=5256.25
total_direct_cost_sum=None arithmetic_check_count=22